



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:16 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:17 PM**Time taken** 1 min**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int find_floor(int *arr, int low, int high, int x) {
5     if (low > high) {
6         return -1;
7     }
8
9     if (x >= arr[high]) {
10        return arr[high];
11    }
12
13    int mid = low + (high - low) / 2;
14
15    if (arr[mid] == x) {
16        return arr[mid];
17    }
18
19    if (mid > 0 && arr[mid - 1] <= x && x < arr[mid]) {
20        return arr[mid - 1];
21    }
22
23    if (x < arr[mid]) {
24        return find_floor(arr, low, mid - 1, x);
25    }
26
27    return find_floor(arr, mid + 1, high, x);
28 }
29
30 int main() {
31     int n;
32     if (scanf("%d", &n) != 1) {
33         return 0;
34     }
35
36     int *arr = (int *)malloc(n * sizeof(int));
37     if (arr == NULL) {
38         return 1;
39     }
40
41     for (int i = 0; i < n; i++) {
42         scanf("%d", &arr[i]);
43     }
44
45     int x;
46     if (scanf("%d", &x) != 1) {
47         free(arr);
48         return 0;
49     }
50
51     int result = find_floor(arr, 0, n - 1, x);
52     printf("%d\n", result);
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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